AEROTWIN SYSTEM ARCHITECTURE

High-Level Design & Data Pipeline Blueprint (Team: Binary-Minds)

LAYER 1: TELEMETRY & INGESTION

CPCB Ground Sensors
Historical & live ambient air quality monitoring (PM2.5, AQI).
`CSV / REST API`

Static EHR Profiles
Patient demographics and comorbidities (Asthma, HTN).
`Synthea Dataset`

Wearable Vitals
Longitudinal health baseline (Resting HR, SpO2, HRV).
`Telemetry JSON`

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LAYER 2: BIOLOGICAL RESPONSE ENGINE

Data Fusion & Validation
Null-value imputation, time-series alignment, and one-hot encoding of clinical variables.
`Pandas Pipeline`

In-Silico Mass-Balance
Calculates Accumulated Internal Dose via EWMA, adjusting for Indoor/Outdoor ratio and N95 masks.
`Cyber-Clinical ODE`

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LAYER 3: PREDICTIVE INFERENCE

XGBoost Ensemble
8-model bootstrap ensemble with monotone constraints predicting 24-hour flare-up probability.
`scikit-learn / joblib`

XAI (Explainable AI)
Extracts SHAP feature contributions to map exact physiological drivers (e.g., Asthma multiplier).
`SHAP TreeExplainer`

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LAYER 4: ACTION & PRESENTATION

Interactive Dashboard
Single-page command center for clinical oversight, data exploration, and visualization.
`Streamlit UI`

What-If Simulator
Interactive toggles (Purifier, Mask) that trigger instant recalculation of predicted risk.
`Plotly Graphs`

External Data · Processing & Math · AI Engine · User Interface